

WEEK 2 RECAP AND EXAM REVIEW

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WEEK 2

WE NOW KNOW A FEW WAYS TO THINK ABOUT THE MONOPOLY'S TRADE-OFF

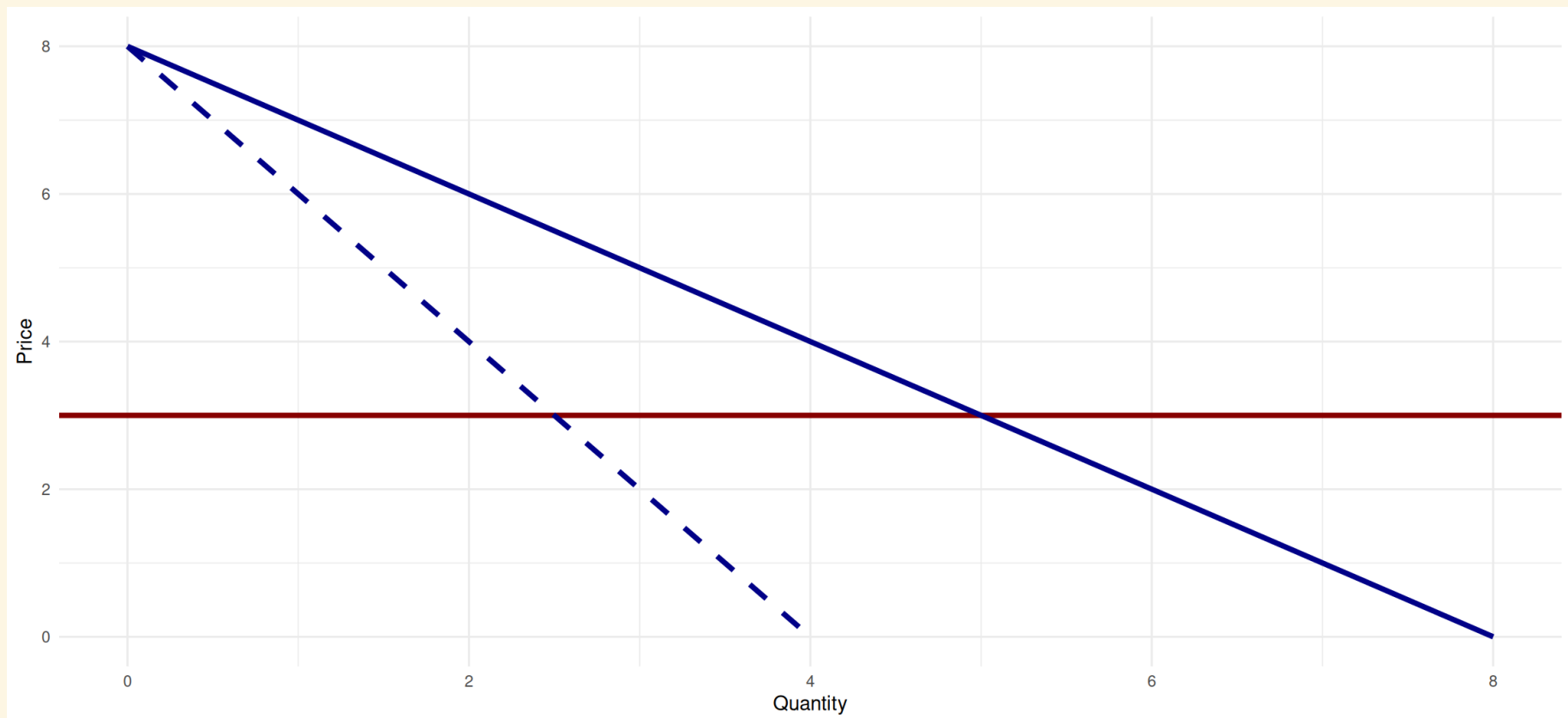
When the monopoly tries to raise prices, it is trading-off:

- Making more margin on the customers it keeps
- Losing customers who are priced out of the market

WE FIRST LEARNT THIS IS EQUIVALENT TO MARGINAL REVENUE AND MARGINAL COST

Profit being maximised is equivalent to marginal revenue intersecting marginal costs:

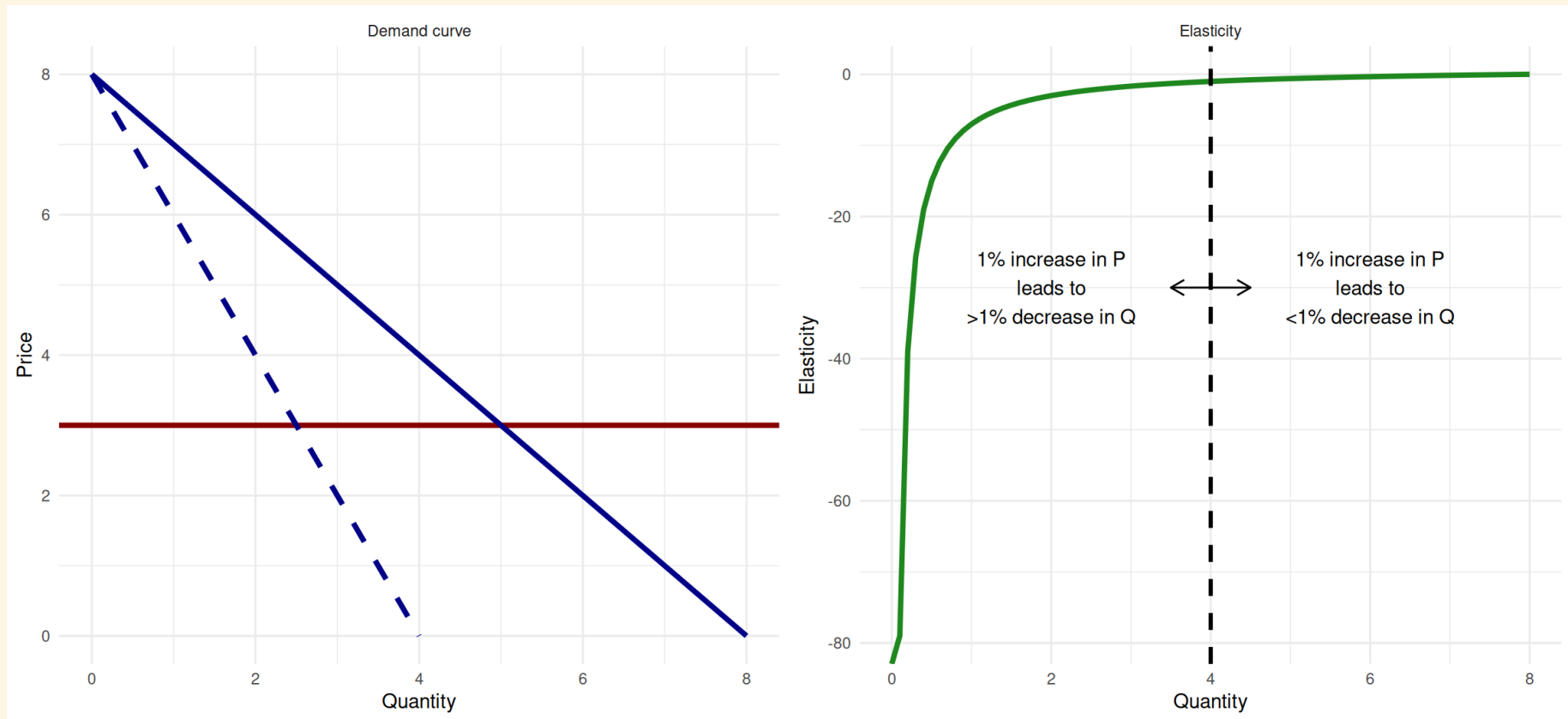
$$\frac{d\pi}{dQ} = 0 \iff MR = MC$$



We see that when $MR > MC$, the monopolist is pricing too high. When $MR < MC$, the

monopolist is setting the price too low

NEXT, THERE'S A CONNECTION BETWEEN THE TRADE-OFF AND CUSTOMER ELASTICITY



TWO THINGS WE LEARN FROM CUSTOMER ELASTICITY

1. Monopolists will always charge prices in the elastic region of customer demand
 - If the monopolist finds itself in the inelastic region of demand, it could always increase prices profitably
2. Monopolists would prefer customers with lower price sensitivity
 - When customers are less elastic (lower price sensitivity) they can increase prices higher without losing as many customers
 - This allows then to increase prices profitably

LASTLY, THE NATURE OF THE BARRIERS TO ENTRY AFFECTS THE POLICY WE USE TO MANAGE MONOPOLIES

Innovation, research, etc:

- If there is a one-time upfront cost that is **large**, we might need to allow a company to be a monopoly for the market to sustain itself
- If there is a one-time upfront cost that is **small**, we would prefer a competitive market

Natural monopolies:

- If upfront costs are large and need to be repeated by every possible competitor, we're best to allow a company to be a monopoly
- We often regulate these firms and set the maximum price at which they can sell

EXAMPLE TIME

Amtrak wants to build a new train line. It has lined up two possible routes:

- Sacramento to Las Vegas - with demand: $Q(1,000 \text{ trips per month}) = 25 - P$
- Sacramento to Nevada City - with demand: $Q(1,000 \text{ trips per month}) = 30 - 2P$

Amtrak has a marginal cost per customer of \$5.

Q (a): If Amtrak can operate as a monopoly, it will sell 10 (thousand) trips per month. Which route should it pick?

EXAMPLE

Q (b): So Amtrak is going to build the new line from Sac to Las Vegas. It's going to cost \$50,000 and it wants it to be paid back within one month.

What's the minimum price it could charge to make its money back?

What is the total surplus (including investment cost)?

EXAMPLE

Q (c): The transport regulator wants to encourage competition in transport and encourages another trainline to compete on this route, requiring it to build a new track.

What is the minimum price that could sustain both train operators, assuming total industry profits are split 50-50?

If the regulator wants to make social surplus as large as possible, should it encourage this competition?

